

Online Zero-Sum Graphical Game for L_2 -Gain Synchronization in Multi-UAV Formation Control

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I. INTRODUCTION

Unmanned Aerial Vehicles (UAVs) have been extensively employed in a wide range of civilian and military applications over the past two decades, owing to their small size, high mobility, low operational cost, and ability to operate in hazardous environments [1,2]. Compared with single-UAV missions, multi-UAV formation flight offers significant advantages in mission efficiency, fault tolerance, and operational flexibility, which has made it a focal research topic in recent years [3–5]. In formation control, each UAV can be regarded as an autonomous decision-maker that cooperates with its neighbors through local information exchange while simultaneously responding to environmental influences such as wind gusts, sudden payload variations, aerodynamic coupling, and actuator faults.

A fundamental requirement in multi-UAV formation control is the design of distributed protocols that depend only on local neighbor information to drive all UAVs toward a common desired trajectory, a problem commonly referred to as consensus or synchronization [6–12]. The literature on distributed consensus is rich, spanning first- and second-order dynamics, leaderless and leader-follower configurations, switching topologies, and time-delay effects [8–12]. Nevertheless, the majority of existing consensus algorithms assume an idealized disturbance-free environment, an assumption that is rarely satisfied in real-world UAV operations where wind turbulence, communication delays, and payload disturbances are unavoidable.

External disturbances can significantly degrade tracking performance and may even destabilize the formation. Consequently, robust consensus against external disturbances has become a critical research direction in multi-agent systems. Early works addressed this issue through H control [13–15], linear matrix inequalities (LMIs) [16,17], or finite L_2 -gain formulations [18,19]. In particular, the bounded L_2 -gain synchronization problem, which seeks to guarantee that the synchronization error is attenuated below a prescribed level with respect to the disturbance energy, has received increasing attention due to its direct physical interpretability and its natural suitability for UAV systems operating in turbulent environments [14,18,19].

Game theory provides a powerful mathematical framework for analyzing the interactions of multiple decision-makers and

has been recognized as an effective tool for addressing optimal control problems in multi-agent systems subject to disturbances [20–22]. In particular, two-player zero-sum differential games, in which the control player minimizes the cost while the disturbance player maximizes it, offer a natural formulation for the bounded L_2 -gain problem [23]. The solution of such games is characterized by the Hamilton–Jacobi–Isaacs (HJI) equation, whose analytical solution is generally intractable for nonlinear systems. To overcome this difficulty, policy iteration (PI) and adaptive dynamic programming (ADP) techniques have been developed to approximate the value function and to compute the Nash equilibrium solution in an iterative manner [24–27]. ADP, first proposed by Werbos [28] for the design of optimal controllers, has since been widely applied to robotics, power systems, and aerospace systems.

In the multi-UAVs domain, ADP-based methods have been investigated for path planning [29], resource allocation [30], cooperative fault-tolerant control [31], and formation control under disturbances [32–34]. Notably, Guan and Fu [34] formulated a discrete-time two-player zero-sum game for multi-UAVs systems and employed a three-layer neural network to approximate the value function, achieving satisfactory tracking performance. However, their formulation is restricted to a two-player zero-sum game in which the disturbance input enters only the dynamics of the controlled UAV, so that the cross-coupling induced by disturbances acting on neighboring UAVs is not captured. Moreover, the discrete-time formulation may not faithfully represent the continuous-time aerodynamic dynamics of UAVs, which are typically described by differential equations.

In the broader multi-agent literature, Jiao, Modares, Xu, Lewis, and Vamvoudakis [35] introduced the concept of differential graphical games, in which each agent’s dynamics and local performance index depend on the policies of its neighbors. This framework leads to coupled Hamilton–Jacobi–Isaacs (HJI) equations whose solution requires the simultaneous update of all agents’ control and disturbance policies. A multi-agent learning policy iteration algorithm was provided together with a rigorous proof of convergence, and it was shown that the global synchronization error is L_2 -bounded when the communication graph contains a spanning tree. An online actor-critic-disturbance implementation was also proposed, in which three neural networks per agent are tuned simultaneously in real time, with Lyapunov-based

convergence guarantees. Although this graphical-game formulation provides a faithful and rigorous treatment of the cross-coupling among neighboring agents, its application to UAV formation systems has not yet been reported in the literature.

Motivated by the above observations, this paper investigates the application of the continuous-time multi-player zero-sum differential graphical game framework of Jiao et al. [35] to the L_2 -gain synchronization problem of multi-UAV formation control. Rather than proposing a new algorithm, the contribution of this paper lies in transferring an established theoretical framework to the UAV domain and demonstrating its practical effectiveness through simulation. Specifically, the bounded L_2 -gain synchronization problem for a multi-UAV formation is formulated as a continuous-time multi-player zero-sum graphical game, in which each UAV's local dynamics and local performance index depend on the states and policies of its neighbors as well as on external disturbances. This formulation faithfully captures the cross-coupling induced by both control actions and disturbances of neighboring UAVs, which is missing from existing two-player formulations such as that in [34]. The coupled Hamilton–Jacobi–Isaacs (HJI) equations characterizing the Nash equilibrium of the proposed graphical game are derived, and the result of [35] is invoked to guarantee that the global synchronization error is L_2 -bounded whenever the communication graph contains a directed spanning tree. The online multi-agent policy iteration algorithm of [35] is then implemented through an actor-critic-disturbance neural-network architecture, with three independent networks per UAV tuned simultaneously in real time, and its convergence to the optimal Nash equilibrium follows from the Lyapunov analysis of [35]. Finally, a comprehensive simulation study on a multi-UAV formation subject to time-varying external disturbances is provided, demonstrating that the algorithm achieves L_2 -bounded synchronization errors in real time without requiring prior knowledge of the disturbance distribution.

The remainder of this paper is organized as follows. Section 2 introduces the graph-theoretic preliminaries and formulates the L_2 -gain synchronization problem. Section 3 casts this problem as a multi-player zero-sum graphical game, derives the coupled HJI equations, presents the Nash equilibrium analysis, and the offline multi-agent policy iteration algorithm. Section 4 develops the online actor-critic-disturbance implementation. Section 5 presents the simulation results on a multi-UAV formation, and Section 6 concludes the paper.

II. PRELIMINARIES AND PROBLEM FORMULATION

This section describes the basic mathematical theory including algebraic graph theory and optimal control theory and formulates an optimal tracking control problem

A. Preliminaries on Graph Theory

The communication topology of a multi-UAVs system is modeled using the graph $G = \{V, E\}$, where it denotes a nonempty finite set $V = \{s_1, s_2, \dots, s_N\}$, the set of edges $E \subseteq VV$, and it defines the connectivity. Matrix E make

$E = [a_{ij}]$ is called the adjacency matrix with $a_{ij} > 0$ if $(s_j, s_i) \in E$ and $a_{ij} = 0$ otherwise. Note that diagonal elements $a_{ii} = 0$.

A node j is said to be a neighbor of a node i , if node i can send a message to node j , and the set of neighbor nodes of i can be denoted as $N_i = \{s_j : (s_j, s_i) \in E\}$. A graph is said to contain a directed spanning tree if there are nodes in the graph that can reach other nodes via directed paths. Define the incidence matrix H as the diagonal matrix $H = \text{diag}(h_i)$, where $h_i = \sum_{j \in N_i} a_{ij}$ denotes the incidence of node i . The graph Laplacian matrix L is defined as $L = H - E$, and the elements of the Laplacian matrix l_{ij} are defined as follows

$$l_{ij} = \begin{cases} -a_{ij}, & i \neq j \\ \sum_{m \in N_i} a_{im}, & i = j \end{cases} \quad (1)$$

In this paper, a system consisting of N UAVs subject to unknown external disturbances is considered. The trajectory tracking problem of the multi-UAVs system is addressed.

Definition 1 (Lewis et al., 2004). A (directed) tree is a connected digraph where every node except one, called the root, has in degree equal to one. A graph is said to have a spanning tree if a subset of the edges forms a directed tree.

Throughout the paper, $\bar{\sigma}(f)$ and $\sigma(f)$ are denoted as the maximum and minimum singular values of the matrix f , respectively

B. Problem formulation

Consider the communication topology of a multi-UAVs is modelled using the graph $G(V, E)$ having N agents, each with dynamics given by

$$\dot{x}_i = f x_i + g_i u_i + k_i d_i \quad (2)$$

where $x_i(t) \in R^n$, $u_i(t) \in R^{m_i}$ and $d_i(t) \in R^{q_i}$ are the state, the control input, and the external disturbance of node i , respectively. The state of the leader UAV node is $x_0(t) \in R^n$, assuming that it satisfies the dynamics

$$\dot{x}_0 = f x_0 \quad (3)$$

Standard Synchronization Problem. Design $u_i(t)$ in (2) so that $\|x_i(t) - x_0(t)\| \rightarrow 0, \forall i$ when $d_i(t) = 0$.

For each UAV, the local neighborhood tracking errors $\delta_i \in R^n$ is defined as Khoo, Xie, and Man (2009)

$$\delta_i = \sum_{j \in N_i} a_{ij} (x_i - x_j) + b_i (x_i - x_0) \quad (4)$$

where $b_i \geq 0$ is called the pinning gain and $b_i > 0$ for least one root node i (Li et al., 2004). The coefficient a_{ij} related to the relationship between UAVs. If there is information exchange between UAV i and UAV j , then $a_{ij} = 1$; otherwise, $a_{ij} = 0$. From (4), the overall tracking error vector for all nodes is given by

$$\delta = ((L + B) \otimes I_n) (x - \bar{x}_0) = ((L + B) \otimes I_n) \xi_i \quad (5)$$

where $x = [x_1 \ x_2 \ \dots \ x_N]^T \in R^{nN}$ is the global node state vector, $\bar{x}_0 = \mathbf{1}_N \otimes x_0 \in R^{nN \times n}$, $\mathbf{1}_N$ is the N -vector of

ones and I_n is the $n \times n$ identity matrix. The Kronecker product is denoted by $\otimes$. The pinning gain matrix $B \in R^{N \times N}$ is a diagonal matrix with diagonal entries equal to the pinning gains b_i . The (global) synchronization error is

$$\xi = x_i - \bar{x}_0 \quad (6)$$

where $\xi = [\xi_1 \ \xi_2 \ \dots \ \xi_N]^T \in R^{nN}$.

The following lemma shows that the small local neighborhood synchronization error implies small global synchronization error.

Lemma 1 (Khoo et al., 2009). *Let $(L + B)$ be non-singular. Then the synchronization error is bounded by*

$$\|\xi\| \leq \|\delta\| / \bar{\sigma} (L + B) \quad (7)$$

Remark 1. The matrix $(L + G)$ is non-singular if the graph has a spanning tree and $b_i \neq 0$ for a root UAV i (Khoo et al., 2009).

To find the dynamics of the local neighborhood tracking error, write

$$\dot{\delta}_i = \sum_{j \in N_i} a_{ij} (\dot{x}_i - \dot{x}_j) + b_i (\dot{x}_i - \dot{x}_0) \quad (8)$$

which by using (2) and (3) becomes

$$\begin{aligned} \dot{\delta}_i = & f_i \delta_i + (h_i + b_i) g_i u_i - \sum_{j \in N_i} a_{ij} g_j u_j + \\ & (h_i + b_i) k_i d_i - \sum_{j \in N_i} a_{ij} k_j d_j \end{aligned} \quad (9)$$

with $\delta_i \in R^n$, $u_i \in R^{m_i}$, $d_i \in R^{q_i}$, $\forall i$. This is a dynamical system with multiple control inputs and disturbance inputs from UAV i and all of her neighbors.

The objective here is to design a distributed control protocol to make the tracking synchronization error (4) L_2 -bounded when $d_i(t) \neq 0$.

Bounded L_2 -Gain Synchronization Problem. Consider the systems (8) with measured outputs $y_i = H_{ii} \delta_i$ (Note that H_{ii} must be left invertible and hence δ_i can be directly measured), disturbances $D_i(t) = [d_i^T(t) \ d_{-i}^T(t)]^T$ and performance outputs $z_i(t) = [\delta_i^T(t) \ u_i^T(t) \ u_{-i}^T(t)]^T$ with $d_{-i}(t) = \{d_j : j \in N_i\}$ and $u_{-i}(t) = \{u_j : j \in N_i\}$. It is desired to select the control $u_i(t)$ to solve the standard synchronization problem when $d_i(t) = 0$, and also to satisfy the following bounded L_2 -gain condition for a given $\gamma > 0$ for all agents when $d_i(t) \neq 0$,

$$\begin{aligned} \int_0^T \|z_i(t)\|^2 dt &= \int_0^T \left(\delta_i^T Q_{ii} \delta_i + u_i^T R_{ii} u_i + \sum_{j \in N_i} u_j^T R_{ij} u_j \right) dt \\ &\leq \gamma^2 \int_0^T \|\omega_i(t)\|^2 dt + \beta(\delta_i(0)) \\ &= \gamma^2 \int_0^T \left(d_i^T T_{ii} d_i + \sum_{j \in N_i} d_j^T T_{ij} d_j \right) dt + \beta(\delta_i(0)) \end{aligned} \quad (10)$$

for some bounded function β such that $\beta(0) = 0$, with $Q_{ii} > 0$, $R_{ii} > 0$, $R_{ij} \geq 0$, $T_{ii} > 0$, $T_{ij} \geq 0$. Let γ^* be the minimum value of γ for which the disturbance attenuation condition is satisfied.

III. MULTI-PLAYER ZERO-SUM DIFFERENTIAL GRAPHICAL GAME AND POLICY ITERATION

Define the following local performance index for each agent.

$$\begin{aligned} J_i(\delta_i(0), u_i, u_{-i}, d_i, d_{-i}) = & \frac{1}{2} \int_0^\infty \left(\delta_i^T Q_{ii} \delta_i + u_i^T R_{ii} u_i \right. \\ & \left. + \sum_{j \in N_i} u_j^T R_{ij} u_j - \gamma^2 d_i^T T_{ii} d_i - \gamma^2 \sum_{j \in N_i} d_j^T T_{ij} d_j \right) dt \end{aligned} \quad (11)$$

A. Two-player zero-sum differential games and Nash equilibrium

Define the following multi-player zero-sum differential game

$$V_i(\delta_i) = \min_{u_i} \max_{d_i} J_i(\delta_i(0), u_i, u_{-i}, d_i, d_{-i}) \quad (12)$$

subject to the dynamical constraints (2). Thus, u is the minimizing player and d is the maximizing one. This game has a unique solution if a game theoretic saddle point (u_i^*, d_i^*) exists, that is, if the Nash condition holds

$$\begin{aligned} V_i^*(\delta_i(0)) &= \min_{u_i} \max_{d_i} J_i(\delta_i(0), u_i, u_{-i}^*, d_i, d_{-i}^*) \\ &= \max_{u_i} \min_{d_i} J_i(\delta_i(0), u_i, u_{-i}^*, d_i, d_{-i}^*) \end{aligned} \quad (13)$$

The associated value V_i^* is called the value of the games. This is equivalent to the Nash equilibrium condition

$$\begin{aligned} J_i(u_i^*, u_{-i}^*, d_i, d_{-i}^*) &\leq J_i(u_i^*, u_{-i}^*, d_i^*, d_{-i}^*) \\ J_i(u_i^*, u_{-i}^*, d_i^*, d_{-i}^*) &\leq J_i(u_i, u_{-i}^*, d_i^*, d_{-i}^*) \end{aligned} \quad (14)$$

holding for all policies u_i, d_i . Interpreting the control inputs $u_i, \forall i$, and the disturbance inputs $d_i, \forall i$ as policies in terms of δ_i , the value function for UAV i is

$$\begin{aligned} V_i(\delta_i(t)) = & \frac{1}{2} \int_t^\infty \left(\delta_i^T Q_{ii} \delta_i + u_i^T R_{ii} u_i + \sum_{j \in N_i} u_j^T R_{ij} u_j \right. \\ & \left. - \gamma^2 d_i^T T_{ii} d_i - \gamma^2 \sum_{j \in N_i} d_j^T T_{ij} d_j \right) dt \end{aligned} \quad (15)$$

Using Leibniz's formula, a differential equivalent to (15) is given by the Bellman equation as

$$\begin{aligned} H_i \left(\delta_i, \frac{\partial V_i}{\partial \delta_i}, u_i, u_{-i}, d_i, d_{-i} \right) \equiv & \frac{\partial V_i^T}{\partial \delta_i} \left(f \delta_i + (h_i + b_i) g_i u_i \right. \\ & \left. - \sum_{j \in N_i} a_{ij} g_j u_j + (h_i + b_i) k_i d_i - \sum_{j \in N_i} a_{ij} k_j d_j \right) \\ & + \frac{1}{2} \delta_i^T Q_{ii} \delta_i + \frac{1}{2} u_i^T R_{ii} u_i + \frac{1}{2} \sum_{j \in N_i} u_j^T R_{ij} u_j \\ & - \frac{1}{2} \gamma^2 d_i^T T_{ii} d_i - \frac{1}{2} \gamma^2 \sum_{j \in N_i} d_j^T T_{ij} d_j = 0 \end{aligned} \quad (16)$$

with the boundary condition $V_i(0) = 0$. The stationarity conditions give (Lewiset al., 2012)

$$0 = \frac{\partial H_i}{\partial u_i} \Rightarrow u_i^* = -(h_i + b_i) R_{ii}^{-1} g_i^T \frac{\partial V_i}{\partial \delta_i} \quad (17)$$

$$0 = \frac{\partial H_i}{\partial d_i} \Rightarrow d_i^* = \frac{1}{\gamma^2} (h_i + b_i) T_{ii}^{-1} k_i^T \frac{\partial V_i}{\partial \delta_i} \quad (18)$$

Remark 2. The game defined by (11), (12) is in fact, a zero-sum game with respect to u_i and d_i for each UAV i , with outcomes coupled through the neighboring relations expressed by the sets N .

B. Coupled Hamilton-Jacobi-Isaacs equations

Substituting control policies (17) and disturbance policies (18) into (16) yields the coupled HJI equations

$$\begin{aligned} & \frac{\partial V_i^T}{\partial \delta_i} f_i^c + \frac{1}{2} \delta_i^T Q_{ii} \delta_i + \frac{1}{2} (h_i + b_i)^2 \frac{\partial V_i^T}{\partial \delta_i} g_i R_{ii}^{-1} g_i^T \frac{\partial V_i}{\partial \delta_i} \\ & + \frac{1}{2} \sum_{j \in N_i} (h_j + b_j)^2 \frac{\partial V_j^T}{\partial \delta_j} g_j R_{jj}^{-1} R_{ij} R_{jj}^{-1} g_j^T \frac{\partial V_j}{\partial \delta_j} \\ & - \frac{1}{2\gamma^2} (h_i + b_i)^2 \frac{\partial V_i^T}{\partial \delta_i} k_i T_{ii}^{-1} k_i^T \frac{\partial V_i}{\partial \delta_i} \\ & - \frac{1}{2\gamma^2} \sum_{j \in N_i} (h_j + b_j)^2 \frac{\partial V_j^T}{\partial \delta_j} k_j T_{jj}^{-1} T_{ij} T_{jj}^{-1} k_j^T \frac{\partial V_j}{\partial \delta_j} = 0 \end{aligned} \quad (19)$$

with the boundary condition $V_i(0) = 0$ where

$$\begin{aligned} f_i^c &= f \delta_i - (h_i + b_i)^2 g_i R_{ii}^{-1} g_i^T \frac{\partial V_i}{\partial \delta_i} + \sum_{j \in N_i} a_{ij} (h_j + b_j) \times \\ & g_j R_{jj}^{-1} g_j^T \frac{\partial V_j}{\partial \delta_j} + \frac{1}{\gamma^2} (h_i + b_i)^2 k_i T_{ii}^{-1} k_i^T \frac{\partial V_i}{\partial \delta_i} \\ & - \frac{1}{\gamma^2} \sum_{j \in N_i} a_{ij} (h_j + b_j) k_j T_{jj}^{-1} k_j^T \frac{\partial V_j}{\partial \delta_j} \end{aligned}$$

is the closed-loop system corresponding to UAV i . For a given solution V_i , define $u_i^* = u_i(V_i)$, $u_{-i}^* = u_{-i}(V_i)$, $d_i^* = d_i(V_i)$ and $d_{-i}^* = d_{-i}(V_i)$, in terms of V_i , then (19) can be written as

$$H_i \left(\delta_i, \frac{\partial V_i}{\partial \delta_i}, u_i^*, u_{-i}^*, d_i^*, d_{-i}^* \right) = 0 \quad V_i(0) = 0 \quad (20)$$

Lemma 2. Assume that the neighboring agents' policies are already optimal. Then, for every u_i , d_i , the following holds

$$\begin{aligned} H_i \left(\delta_i, \frac{\partial V_i}{\partial \delta_i}, u_i^*, u_{-i}^*, d_i^*, d_{-i}^* \right) &= \frac{1}{2} (u_i - u_i^*)^T R_{ii} (u_i - u_i^*) \\ &- \frac{1}{2} \gamma^2 (d_i - d_i^*)^T T_{ii} (d_i - d_i^*) \end{aligned} \quad (21)$$

Theorem 1. (Solution to Bounded L_2 -gain Problems). Let $\gamma \geq \gamma^*$. Suppose $V_i^* > 0$, $i \in \mathcal{N}$ is a smooth positive definite solution to the HJI equation (19). Assume that the neighboring

agents' policies are already optimal. Then, the equilibrium point of the closed-loop system

$$\dot{\delta}_i = f \delta_i + (h_i + b_i) B_i u_i^* - \sum_{j \in N_i} a_{ij} g_j u_j^* \quad (22)$$

is asymptotically stable with control inputs $u_i^* = u_i(V_i^*)$, given by (17) in terms of V_i^* . Moreover, for these choices of control inputs, the bounded L_2 -gain condition (10) holds for all disturbances $d_i(t) \in L_2[0, \infty)$, where $Q_{ii} = S_{ii}^T S_{ii}$.

Theorem 2. (Solution to Multi-player Zero-sum Game). Let $\gamma \geq \gamma^*$. Assume that the game (12) has a finite value and the neighboring agents' policies are already optimal. Suppose $V_i^* > 0$, $i \in \mathcal{N}$ is a smooth positive definite solution to the HJI equation (19) such that the equilibrium point of the closed-loop system

$$\begin{aligned} \dot{\delta}_i &= f \delta_i + (h_i + b_i) g_i u_i^* - \sum_{j \in N_i} a_{ij} g_j u_j^* \\ &+ (h_i + b_i) k_i d_i^* - \sum_{j \in N_i} a_{ij} k_j d_j^* \end{aligned} \quad (23)$$

is asymptotically stable. Then, the Nash condition (14) is satisfied for control policy $u_i^* = u_i(V_i^*)$ and disturbance policy $d_i^* = d_i(V_i^*)$ as (17) and (18), given in terms of V_i^* . Moreover, the value of the game is given by the HJI solution $V_i^*(\delta_i(0))$.

C. Policy iteration solution of the Hamilton-Jacobi-Isaacs equation

Definition 2. Given fixed neighbor control policies u_{-i} and disturbance policies d_{-i} , assume that there exist stabilizing control policy u_i and disturbance policy d_i . Then, UAV i 's best control and disturbance response to fixed policies u_{-i} and d_{-i} of her neighbors are the pair of policies u_i^* and d_i^* such that (14)

Define the best response HJI equation with control $u_i = u_i^*$ given by (17) and disturbances $d_i = d_i^*$ given by (18) and arbitrary policies u_{-i} , d_{-i} as

$$\begin{aligned} & \frac{\partial V_i^T}{\partial \delta_i} f_i^c + \frac{1}{2} \delta_i^T Q_{ii} \delta_i + \frac{1}{2} (h_i + b_i)^2 \frac{\partial V_i^T}{\partial \delta_i} g_i R_{ii}^{-1} g_i^T \frac{\partial V_i}{\partial \delta_i} \\ & + \frac{1}{2} \sum_{j \in N_i} u_j^T R_{ij} u_j - \frac{1}{2\gamma^2} (h_i + b_i)^2 \frac{\partial V_i^T}{\partial \delta_i} k_i T_{ii}^{-1} k_i^T \frac{\partial V_i}{\partial \delta_i} \\ & - \frac{1}{2} \gamma^2 \sum_{j \in N_i} d_j^T T_{ij} d_j = 0 \end{aligned} \quad (24)$$

with boundary condition $V_i(0) = 0$, where f_i^c , $\forall i$ is the closed-loop system.

A policy iteration (PI) algorithm is presented to solve the best response HJI equations (24) and the HJI equations (19). The PI technique has two loops, i.e. an inner loop (policy evaluation) and an outer loop (policy improvement). In the inner loop, the control policies are fixed and the disturbances are updated using (18) by solving the Bellman equation (16). In the outer loop, improved control policies are found using (17). This gives the following PI algorithm.

Algorithm 1 Multi-agent Learning Policy Iteration for Solving HJI equations

Let $u_i^0, \forall i = 1, \dots, N$ be any stabilizing initial policy.
for $k = 0, 1, \dots$ **do**, given $u_i^k, \forall i = 1, \dots, N$.
 repeat at each step k
 for $l = 0, 1, \dots$ **do**
 Set $d_i^0 = 0$, (Policy evaluation)
 repeat at each step l
 Solve for $d_i^{(k,l)}(\delta_i)$ using the zero-sum Bellman equation:

$$\begin{aligned} & \left(\frac{\partial V_i^{(k,l)}}{\partial \delta_i} \right)^T \left(f\delta_i + (h_i + b_i)g_i u_i^k - \sum_{j \in N_i} a_{ij}g_j u_j \right. \\ & + (h_i + b_i)k_i d_i^l - \sum_{j \in N_i} a_{ij}k_j d_j \left. \right) + \frac{1}{2} \delta_i^T Q_i \delta_i \\ & + \frac{1}{2} u_i^{kT} R_{ii} u_i^k + \frac{1}{2} \sum_{j \in N_i} u_j^{kT} R_{ij} u_j - \frac{1}{2} \gamma^2 d_i^{lT} T_{ii} v_i^l \\ & - \frac{1}{2} \gamma^2 \sum_{j \in N_i} d_j^{lT} T_{ij} d_j = 0, \quad \forall i = 1, \dots, N \end{aligned} \quad (25)$$

Update the disturbances:

$$d_i^{l+1} = \frac{1}{\gamma^2} (h_i + b_i) T_{ii}^{-1} k_i^T \frac{\partial V_i^{(k,l)}}{\partial \delta_i}, \quad \forall i = 1, \dots, N \quad (26)$$

until convergence

Set $V_i^{(k+1,l)} = V_i^{(k,l)}$

end for

Update the control policy (Policy improvement):

$$u_i^{k+1} = -(h_i + b_i) R_{ii}^{-1} g_i^T \frac{\partial V_i^{(k,l)}}{\partial \delta_i}, \quad \forall i = 1, \dots, N \quad (27)$$

until convergence

end for

The following two theorems prove the convergence of the policy iteration Algorithm 1 for the best response HJI equation (24) and the coupled HJI equation (19), respectively. In Theorem 3, only UAV i updates her policy and in Theorem 4, all the UAVs update their policies simultaneously.

Theorem 3. Assume that the policies in the neighborhood are fixed and that there exist a stabilizing control policy and disturbance policy with $\gamma \geq \gamma^*$. Assume that agent i performs Algorithm 1. Then the algorithm converges to the best response u_i and d_i to policies u_{-i} and d_{-i} of the neighbors, and the value converges to the optimal game value $V_i^{(k,l)} \rightarrow V_i^*$, which V_i^* is the solution to the best response HJI equation (24).

Theorem 4. Assume that the policies in the neighborhood are fixed and that there exist a stabilizing control policy and disturbance policy with $\gamma \geq \gamma^*$. Assume that agent i performs Algorithm 1. Then, for small weight a_{ij} , and relative weights β_{ij} , ρ_{ij} , the policies u_i , u_j , d_i , d_j converge to the Nash

equilibrium, and the game values converge to the optimal game value $V_i^{(k,l)} \rightarrow V_i^*$, which V_i^* is the solution to the best response HJI equation (24).

IV. ONLINE POLICY ITERATION FOR LEARNING THE SOLUTION TO THE HJI EQUATIONS IN REAL TIME

In this section, the solution to the HJI equations (19) is learned online in real time based on policy iteration Algorithm 1. An actor critic-disturbance architecture is used to approximate the solution to these HJI equation. Specifically, each UAV has three neural networks (NNs): one critic, one control actor and one disturbance. That is, for N agents, N critic neural networks (NNs) are used to approximate the value functions in the Bellman equation (16), N actor NNs are used to approximate the control policies (27) and N disturbance NNs are used to approximate the disturbance policies (26). All these 3NNs are tuned simultaneously and in realtime.

A. Critic neural network

According to the Weierstrass higher-order approximation Theorem, there exist NNs such that the smooth value functions V_i are approximated using critic NNs as

$$V_i(\delta_i) = W_i^T \phi_i(\delta_i) + \varepsilon_i \quad (28)$$

where W_i are the critic NN weights, $\phi_i \in R^c$ are the critic NN activation function vectors, with c the number of neurons in the hidden layer of the critic NNs. According to the Weierstrass Theorem, the NN approximation error ε_i converges to zero uniformly as $c \rightarrow \infty$. Assuming current weight estimates $\hat{W}_i$, the outputs of the critic NNs are given by

$$\hat{V}_i = \hat{W}_i^T \phi_i \quad (29)$$

Then, the Bellman equation (25) can be approximated at each step as

$$\begin{aligned} H_i(\delta_i, \hat{W}_i, u_i, u_{-i}) &= \frac{1}{2} \delta_i^T Q_{ii} \delta_i + \frac{1}{2} u_i^T R_{ii} u_i + \frac{1}{2} \sum_{j \in N_i} u_j^T R_{ij} u_j \\ &- \frac{1}{2} \gamma^2 d_i^T T_{ii} d_i - \frac{1}{2} \gamma^2 \sum_{j \in N_i} d_j^T T_{ij} d_j \\ &+ \hat{W}_i^T \frac{\partial \phi_i}{\partial \delta_i} (f\delta_i + (h_i + b_i)g_i u_i \\ &- \sum_{j \in N_i} a_{ij}g_j u_j + (h_i + b_i)k_i d_i - \sum_{j \in N_i} a_{ij}k_j d_j = e_{H_i} \end{aligned} \quad (30)$$

It is desired to select $\hat{W}_i$ to minimize the square residual error

$$E_1 = \frac{1}{2} e_{H_i}^2 \quad (31)$$

Then $\hat{W}_i \rightarrow W_i$ which solves (30) in a least-squares sense and e_{H_i} becomes small. Theorem 5 gives a tuning law for the critic weights that minimizes (30).

B. Actor and disturbance neural networks

Based on the control update policies (17), the control policies are given as actor neural networks (Vamvoudakis et al., 2012) as follows,

$$\hat{u}_i \equiv \hat{u}_{i+N} = -(h_i + b_i) R_{ii}^{-1} g_i^T \frac{\partial \phi_i^T}{\partial \delta_i} \hat{W}_{i+N} \quad (32)$$

where $\hat{W}_{i+N}$ denote the current estimated values of the ideal critic NN weights W_i . Moreover, based on the disturbance update policies (18), the disturbance neural networks are chosen as

$$\hat{d}_i \equiv \hat{d}_{i+2N} = \frac{1}{\gamma^2} (h_i + b_i) T_{ii}^{-1} k_i^T \frac{\partial \phi_i^T}{\partial \delta_i} \hat{W}_{i+2N} \quad (33)$$

where $\hat{W}_{i+2N}$ denote the current estimated values of the ideal critic NN weights W_i . The notation $\hat{u}_{i+N}$ and $\hat{d}_{i+2N}$ in (32) and (33) are used to keep indices straight in the proof (the first N weights are used for the critics, the second N weights are for the actors and the rest for the disturbances).

The following two theorems provide the tuning laws for the actor, critic and disturbance NNs that guarantee convergence of the synchronous online multiple agent learning PI algorithm in real time to the game saddle point solution, while guaranteeing closed-loop stability. Define the critic, actor and disturbance NN estimation errors as $\tilde{W}_i = W_i - \hat{W}_i$, $\tilde{W}_{i+N} = W_{i+N} - \hat{W}_{i+N}$ and $\tilde{W}_{i+2N} = W_{i+2N} - \hat{W}_{i+2N}$.

Theorem 5. (Online Zero-Sum Graphical games). *Let the error dynamics be given by (9), and consider the zero-sum differential graphical game formulation in (12). Let the critic NN at each node be given by (29), the control input be given by actor NN (32), and the disturbance input be given by disturbance NN (33). Let tuning for the i th critic NN be provided by*

$$\begin{aligned} \dot{\tilde{W}}_i = & -a_i \frac{\partial E_1}{\partial \tilde{W}_i} = -a_i \frac{\sigma_{i+N}}{(1 + \sigma_{i+N}^T \sigma_{i+N})^2} \left[\sigma_{i+N}^T \hat{W}_i \right. \\ & + \frac{1}{2} \delta_i^T Q_{ii} \delta_i + \frac{1}{2} (h_i + b_i)^2 \hat{W}_{i+N}^T \bar{G}_i \hat{W}_{i+N} \\ & - \frac{1}{2\gamma^2} (h_i + b_i)^2 \hat{W}_{i+2N}^T \bar{K}_i \hat{W}_{i+2N} \\ & + \frac{1}{2} \sum_{j \in N_i} (h_j + b_j)^2 \hat{W}_{j+N}^T \frac{\partial \phi_j}{\partial \delta_j} g_j R_{jj}^{-T} R_{ij} R_{jj}^{-1} \\ & \times g_j^T \frac{\partial \phi_j^T}{\partial \delta_j} \hat{W}_{j+N} - \frac{1}{2\gamma^2} \sum_{j \in N_i} (h_j + b_j)^2 \hat{W}_{j+2N}^T \\ & \times \left. \frac{\partial \phi_j}{\partial \delta_j} k_j T_{jj}^{-T} T_{ij} T_{jj}^{-1} k_j^T \frac{\partial \phi_j^T}{\partial \delta_j} \hat{W}_{j+2N} \right] \end{aligned} \quad (34)$$

where

$$\begin{aligned} \sigma_{i+N} = & \frac{\partial \phi_i}{\partial \delta_i} \left(f \delta_i + (h_i + b_i) g_i \hat{u}_{i+N} - \sum_{j \in N_i} a_{ij} g_j \hat{u}_{j+N} \right. \\ & \left. + (h_i + b_i) k_i \hat{d}_{i+2N} - \sum_{j \in N_i} a_{ij} k_j \hat{d}_{j+2N} \right) \end{aligned} \quad (35)$$

Let the i th actor NN be tuned as

$$\begin{aligned} \dot{\tilde{W}}_{i+N} = & -a_{i+N} \left\{ (S_i \hat{W}_{i+N} - \bar{F}_i \sigma_{i+N}^T \hat{W}_i) - (h_i + b_i)^2 \bar{G}_i \right. \\ & \times \hat{W}_{i+N} \frac{\bar{\sigma}_{i+N}^T}{m_{si}} \hat{W}_i - \sum_{j \in N_i} (h_j + b_j)^2 \frac{\partial \phi_j}{\partial \delta_j} g_j R_{ij}^{-T} R_{ij} \\ & \times \left. R_{ij}^{-1} g_j^T \frac{\partial \phi_j^T}{\partial \delta_j} \hat{W}_{i+N} \frac{\bar{\sigma}_{i+N}^T}{m_{si}} \hat{W}_j \right\} \end{aligned} \quad (36)$$

and the i th disturbance NN be tuned as

$$\begin{aligned} \dot{\tilde{W}}_{i+2N} = & -a_{i+2N} \left\{ (\bar{S}_i \hat{W}_{i+2N} - \bar{F}_i \bar{\sigma}_{i+N}^T \hat{W}_i) + \frac{1}{\gamma^2} (h_i + b_i)^2 \bar{K}_i \right. \\ & \times \hat{W}_{i+N} \frac{\bar{\sigma}_{i+N}^T}{m_{si}} \hat{W}_i + \frac{1}{\gamma^2} \sum_{j \in N_i} (h_j + b_j)^2 \frac{\partial \phi_j}{\partial \delta_j} k_j T_{ij}^{-T} T_{ij} T_{ij}^{-1} \\ & \times \left. k_j^T \frac{\partial \phi_j^T}{\partial \delta_j} \hat{W}_{i+2N} \frac{\bar{\sigma}_{i+N}^T}{m_{si}} \hat{W}_j \right\} \end{aligned} \quad (37)$$

where

$$\begin{aligned} \bar{G}_i(x) &= \frac{\partial \phi_i}{\partial \delta_i} g_i R_{ii}^{-1} g_i^T \frac{\partial \phi_i^T}{\partial \delta_i} \\ \bar{K}_i(x) &= \frac{\partial \phi_i}{\partial \delta_i} k_i T_{ii}^{-1} k_i^T \frac{\partial \phi_i^T}{\partial \delta_i} \end{aligned}$$

and $m_{si} \equiv (\sigma_{i+N}^T \sigma_{i+N} + 1)$, $\bar{\sigma}_{i+N} = \frac{\sigma_{i+N}}{\sigma_{i+N}^T \sigma_{i+N} + 1}$ and $a_i > 0, \dots, a_{i+N} > 0, \dots, a_{i+2N} > 0, \bar{F}_i > 0, S_i > 0, \bar{F}_i > 0, \bar{S}_i > 0, i \in N$ are tuning parameters. Suppose that $\bar{\sigma}_{i+N} = \sigma_{i+N} / (\sigma_{i+N}^T \sigma_{i+N} + 1)$ is persistently exciting. Then the closed-loop system states $\delta_i(t)$, the critic NN errors $\tilde{W}_i$, the actor NN errors $\tilde{W}_{i+N}$ and the disturbance NN errors $\tilde{W}_{i+2N}$ are uniformly ultimately bounded.

Theorem 6. (Convergence to Zero-Sum Nash Equilibrium). *Suppose the hypotheses of Theorem 5 hold. Then:*

- 1) $H_i(\delta_i, \tilde{W}_i, \hat{u}_i, \hat{u}_{-i}, \hat{d}_i, \hat{d}_{-i}), \forall i \in N$ are uniformly ultimately bounded, that is, $\tilde{W}_i$ converge to the approximate coupled HJI solution, the value of the ZS game, where

$$\hat{u}_i = -(h_i + b_i) R_{ii}^{-1} g_i^T \frac{\partial \phi_i^T}{\partial \delta_i} \tilde{W}_i,$$

$$\hat{d}_i = \frac{1}{\gamma^2} (h_i + b_i) T_{ii}^{-1} k_i^T \frac{\partial \phi_i^T}{\partial \delta_i} \tilde{W}_i.$$

- 2) $\hat{u}_{i+N}, \hat{d}_{i+2N}$ (see (32) and (33)) converge to the approximate Nash equilibrium solution of the zero-sum game.

V. SIMULATION RESULTS

In this section, the number of follower UAVs is set as $N = 4$, the total time $T = 300s$, and the time step $t = 0.001s$. The communication topology structure of the four follower

$$\text{UAVs is } A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}, H = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}, B =$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$
 The performance index weight matrices are $Q_{ii} = 1, R_{ii} = 1, T_{ii} = 1, Q_{ij} = 1, R_{ij} = 1, T_{ij} = 1$ and the coefficient $\gamma = 1$.

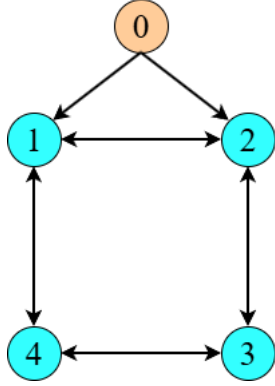


Fig. 1. Communication graph

For the digraph structure 1 shown, we consider the following UAV dynamics

$$\dot{x}_i = f x_i + g_i u_i + k_i d_i$$

where

$$\begin{aligned}
 A &= \begin{bmatrix} 0 & 0.1 \\ -0.1 & 0 \end{bmatrix}, \quad B_1 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad B_2 = \begin{bmatrix} 2 \\ -3 \end{bmatrix}, \\
 B_3 &= \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \quad B_4 = \begin{bmatrix} 1 \\ -3 \end{bmatrix}, \quad D_1 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \\
 D_2 &= \begin{bmatrix} 2 \\ -2 \end{bmatrix}, \quad D_3 = \begin{bmatrix} 2 \\ -2 \end{bmatrix}, \quad D_4 = \begin{bmatrix} 2 \\ -3 \end{bmatrix},
 \end{aligned}$$

Persistence of excitation was ensured by adding a small exponentially decreasing probing noise to the control inputs. The dynamics of each UAV is marginally stable and so synchronization is required to a Lissajous pattern in the phase plane.

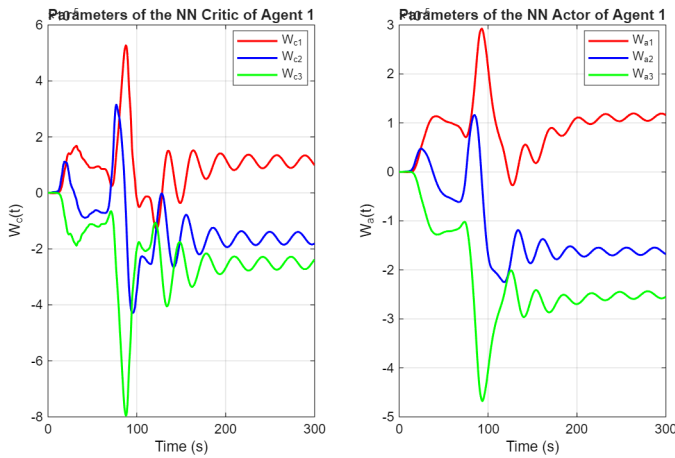


Fig. 2. Parameters of the NN Actor-Critic Agent 1

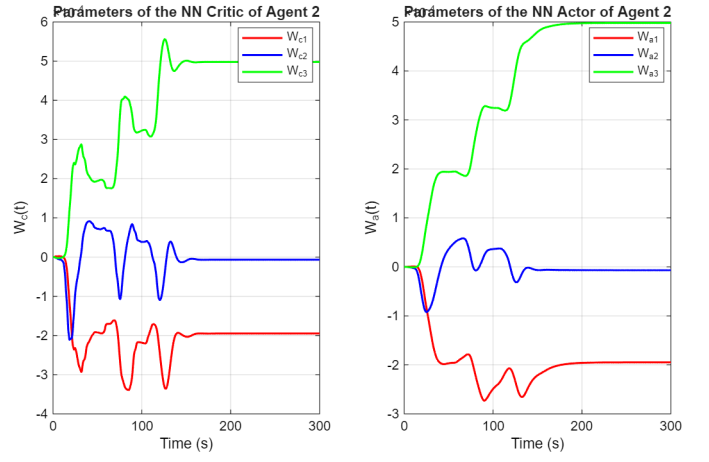


Fig. 3. Parameters of the NN Actor-Critic Agent 2

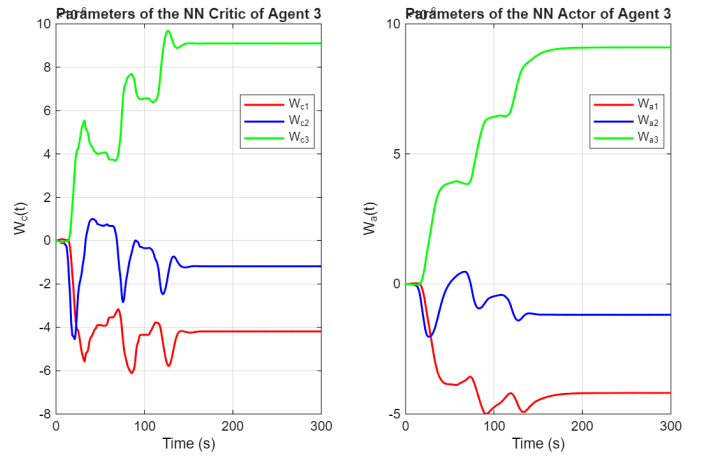


Fig. 4. Parameters of the NN Actor-Critic Agent 3

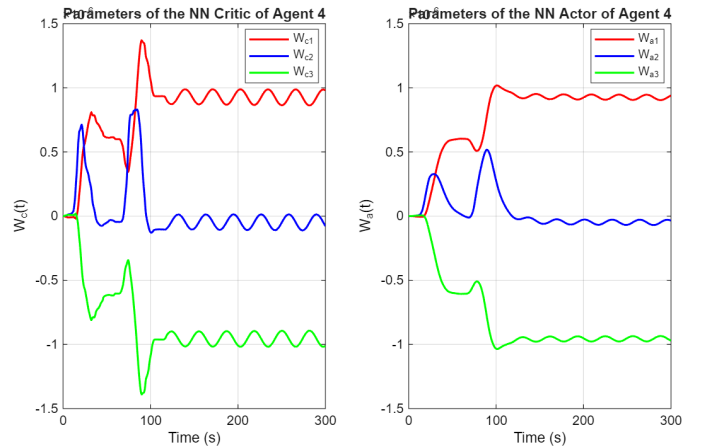


Fig. 5. Parameters of the NN Actor-Critic Agent 4

The convergence of neural network weights in multi-UAV systems is shown in (2),(3),(4) and (5) where the parameters converge after 100s with. After converging, the PE noise can be canceled anytime, for example, here is after 200s.

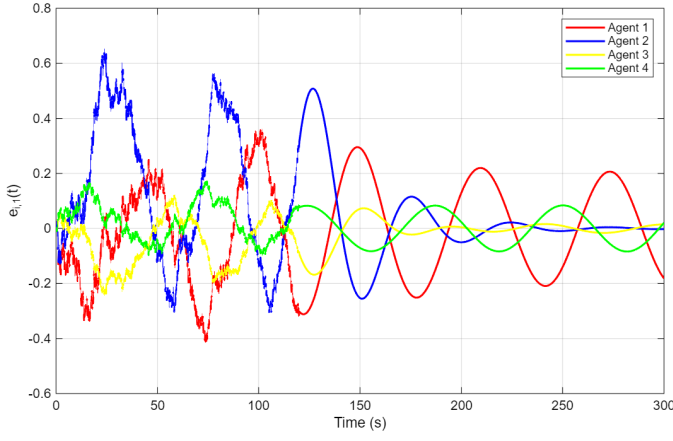


Fig. 6. Evolution of the states for $\delta_{11}, \delta_{21}, \delta_{31}, \delta_{41}$

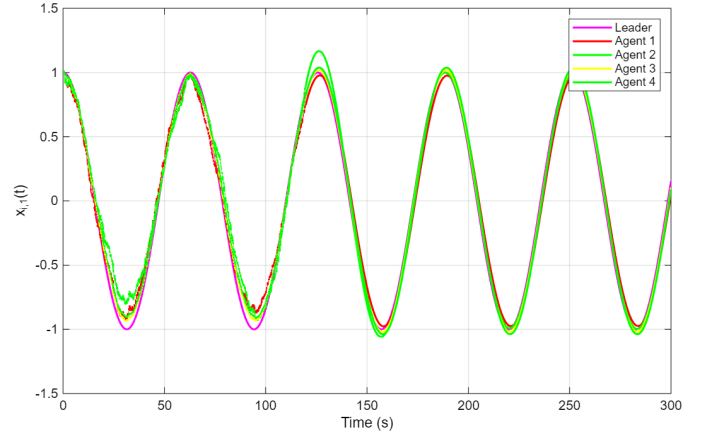


Fig. 8. Evolution of the states for $x_{11}, x_{21}, x_{31}, x_{41}, x_{01}$

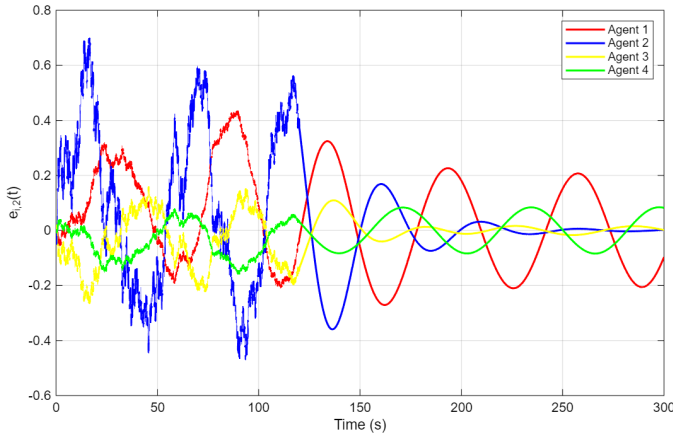


Fig. 7. Evolution of the states for $\delta_{12}, \delta_{22}, \delta_{32}, \delta_{42}$

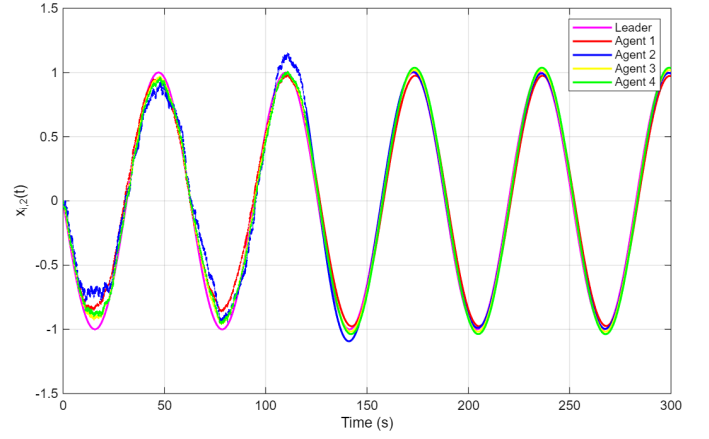


Fig. 9. Evolution of the states for $x_{12}, x_{22}, x_{32}, x_{42}, x_{02}$

At the initial stage, the position error of the distributed UAV system is relatively large. However, owing to the sustainable adaptive optimization capability, the distributed multi-UAVs network progressively reduces the error values while minimizing the energy consumption during the motion of each individual agent. After the neural networks of the UAVs converge, Figure (6) and (7) demonstrate that the system error gradually decreases and stabilizes at a small value close to zero (approximately 0.2). Furthermore, Figure (8) and (9) show that, once the exploration noise is removed, the distributed UAV system is able to maintain formation and follow the prescribed trajectory despite the presence of model disturbances.

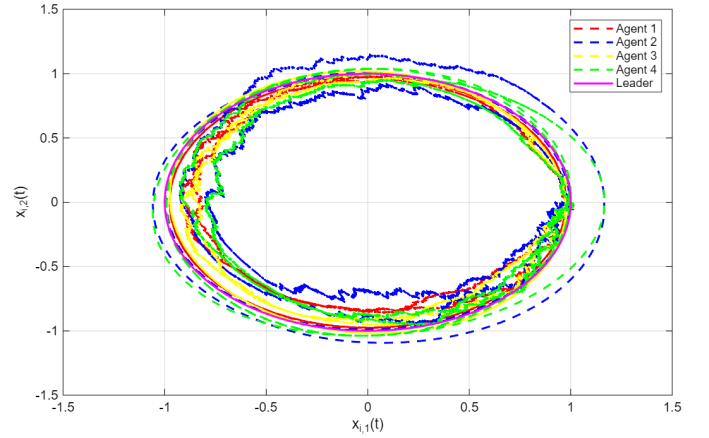


Fig. 10. 2-D phase plane plot

Finally, Figure (10), (11) illustrates the 2D and 3D trajectories of the UAVs under online ADP via Zero-Sum Graphical Game control. The follower UAVs closely track the leader's reference trajectory, with maximum deviations of 0.2 units along both the x_1 - and x_2 -axes. Although the

follower UAVs generally maintain satisfactory alignment with the leader, noticeable offsets remain, as previously discussed. In particular, UAV 1 (red) exhibits the largest deviation, which is clearly observable in the 2D plots. These results highlight the capability of the proposed algorithm to achieve accurate trajectory tracking in complex 2D and 3D environments.

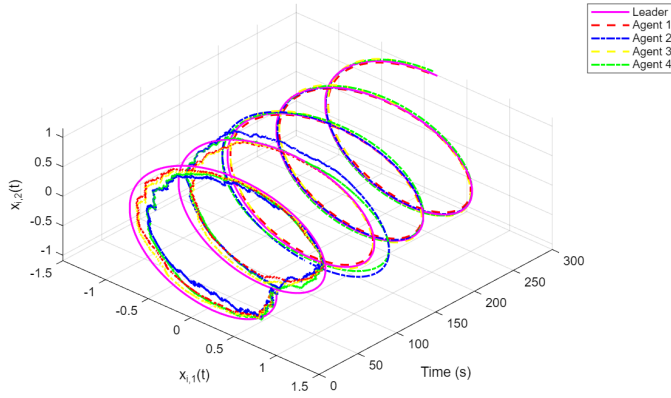


Fig. 11. 3-D phase plane plot

However, the selection of exploration noise still exerts a significant influence on the convergence of the distributed multi-UAV system's weights. Exploration noise may cause the weights to converge to local extrema rather than the global optimum. Therefore, additional time is required to refine the simulations and to compare the obtained results with previous findings in order to verify whether the achieved values are indeed optimal. In summary, the proposed algorithm demonstrates clear advantages in terms of tracking accuracy, control stability, and robustness against disturbances for distributed multi-UAV systems.

VI. CONCLUSION

This paper has presented an application of the continuous-time multi-player zero-sum differential graphical game framework of Jiao et al. [35] to the L_2 -gain synchronization problem of multi-UAV formation control. The bounded L_2 -gain synchronization problem has been formulated as a continuous-time multi-player zero-sum graphical game, the coupled Hamilton–Jacobi–Isaacs (HJI) equations have been derived, and the online multi-agent policy iteration algorithm of [35] has been re-implemented through an actor-critic-disturbance neural-network architecture. Simulation results on a four-UAV formation subject to time-varying external disturbances have demonstrated that the algorithm achieves L_2 -bounded synchronization errors in real time without requiring prior knowledge of the disturbance distribution.

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